

Contents

- [Project 7 report](#)
- [step 1 - evaluate color accuracy of an un-calibrated imaging workflow](#)
- [step 2 - reproduce the submitted calibrated-workflow evaluation](#)
- [step 3 - visualize the differences between ground-truth, un-calibrated and calibrated renderings of CC chart](#)
- [step 4 - color-accurate imaging](#)
- [compare the original and color calibrated images](#)

Project 7 report

zl 12/1/2024

This project used tools developed in earlier projects to analyze the sources of error in color imaging and perform end-to-end color-accurate colorimetric imaging using the camera and display models developed in previous projects.

First, the color differences between the un-calibrated (camera image -> display) and the calibrated (camera image -> camera model -> estimated XYZs -> display model -> color-calibrated RGBs for the given display) imaging workflows were measured and compared. Then, a figure that visualizes these differences were created. Finally, 2 functions, camRGB2XYZ and XYZ2dispRGB, were used to process the ColorChecker image captured in project 1, so the colors were rendered correctly for the given display.

```
repo_dir = fileparts(fileparts(fileparts(fileparts(mfilename('fullpath')))));
addpath(fullfile(repo_dir, 'shared'));
paths = imgs351Paths(repo_dir);
addpath(paths.p03.code, paths.p04.code, paths.p05.code, paths.p06.code);
```

step 1 - evaluate color accuracy of an un-calibrated imaging workflow

```
% load the 8-bit RGB DCs of the ColorChecker patches extracted from camera
% image
camRGBs_255 = load(fullfile(paths.p07.data, 'camRGBs.mat'));
camRGBs_255 = camRGBs_255.cam_RGBs';

% workflow_test_uncal.tif stores these camera RGB stimuli plus repeated
% display-black and display-white measurements in ArgyllCMS format.

% use the ColorMunki colorimeter to measure the XYZs of the camera RGBs displayed on
% the given display without using calibrated models

% load the measured XYZs
uncal_XYZs = importdata(fullfile(paths.p07.data, ...
    'workflow_test_uncal.tif'), ' ', 20);
% extract the XYZ data for the displayed CC patches
uncal_CC_XYZs = uncal_XYZs.data(1:24, 5:7)';
% for 3 measurements of display black and average them
uncal_XYZk = mean(uncal_XYZs.data(25:27, 5:7), 1)';
% for 3 measurements of display white and average them
uncal_XYZw = mean(uncal_XYZs.data(28:30, 5:7), 1)';

% calculate Lab values for the displayed CC patches from the XYZs
uncal_CC_Labs = XYZ2Lab(uncal_CC_XYZs, uncal_XYZw);

% load the ColorMunki measured XYZ and Lab values of the physical CC chart
% into 2 arrays for XYZ and Lab values respectively
munki_data = importdata(fullfile(paths.p07.data, ...
    'munki_CC_XYZs_Labs.txt'));
munki_XYZs = munki_data(:, 2:4)';
munki_Labs = munki_data(:, 5:end)';

% calculate the delta E to measure the color differences
uncal_deltaE = deltaEab(munki_Labs, uncal_CC_Labs);

% calculate the min, max and mean delta E values
uncal_deltaE_max = max(uncal_deltaE);
uncal_deltaE_min = min(uncal_deltaE);
uncal_deltaE_mean = mean(uncal_deltaE);

% summarize and display the differences between the real and displayed Lab
% values for the un-calibrated workflow
printWorkflowError('Uncalibrated', 'camera -> RGB_cam -> display', ...
    munki_Labs, uncal_CC_Labs, uncal_deltaE);

% delete temporary vars
clear munki_data uncal_XYZs
```

Uncalibrated workflow color error
camera -> RGB_cam -> display

Real vs. displayed ColorChecker Lab values

patch #	real			displayed			dEab
	L	a	b	L	a	b	
1	37.1865	14.9985	15.2592	27.6596	16.3892	19.6667	10.5887
2	65.8188	16.8695	18.0267	71.8118	17.9897	18.8794	6.1562
3	49.9949	-3.1841	-23.5159	45.2304	4.3805	-27.6682	9.8573
4	42.6411	-15.3251	20.0423	34.3359	-18.9765	25.2164	10.4442
5	54.6852	9.6978	-26.7126	53.2127	15.1762	-27.8545	5.7867
6	71.2441	-33.1391	-0.5010	74.3650	-29.3684	0.0607	4.9268
7	62.2558	34.1094	57.7774	64.9012	32.5343	58.1052	3.0962
8	39.5890	9.9980	-43.6388	27.0389	27.5371	-55.2024	24.4712
9	51.8424	48.1403	16.0636	52.9590	50.3033	20.8222	5.3451
10	29.4495	22.4255	-21.7661	13.5034	26.0401	-23.8046	16.4773
11	71.6264	-24.3441	57.6850	78.9127	-28.2918	59.8763	8.5718
12	72.2288	20.6039	69.0149	77.3925	11.1112	61.2733	13.2932
13	28.6402	18.5907	-51.4092	10.0559	40.6433	-56.2601	29.2442
14	54.6309	-39.5493	32.8341	48.2291	-39.4169	33.2711	6.4180
15	42.5988	54.6049	25.7315	36.5585	56.0653	38.0650	13.8106
16	82.4265	3.8689	78.8570	89.5819	-6.3723	70.8313	14.8490
17	51.5476	49.5154	-14.3758	47.6321	55.4124	-12.5703	7.3052
18	49.3892	-26.5473	-28.6645	41.0461	-7.7580	-32.3739	20.8903
19	95.4458	-0.4414	0.0244	92.8353	0.5020	2.5347	3.7425
20	80.0339	0.1309	-0.9345	80.7404	0.0349	2.1549	3.1706
21	66.0107	-0.0004	-1.1463	59.6327	0.9215	0.5824	6.6721
22	50.5546	-0.6207	-0.9616	39.7006	-0.2840	1.0689	11.0474
23	35.1532	-0.0632	-0.9708	16.6781	-2.1348	-1.0155	18.5910
24	20.3224	-0.2858	-0.5603	1.0872	0.2297	-1.0338	19.2480
min							3.0962
max							29.2442
mean							11.4168

step 2 - reproduce the submitted calibrated-workflow evaluation

The saved .tif measurements below were collected using the originally submitted RGB stimuli. The resulting Delta E values are historical results; physically validating the corrected camera/display conversion would require a new display-measurement session.

```
% load the 8-bit RGB DCs of the ColorChecker patches
camRGBs_255;

% put the RGB DCs through camRGB2XYZ function to calculate the XYZs of the
% patches
camXYZs = camRGB2XYZ(fullfile(paths.p05.data, 'cam_model.mat'), ...
    camRGBs_255);

% load the CIE data, calculate XYZs for D50
cie = loadCIEdata;
XYZ_D50 = ref2XYZ(cie.PRD, cie.cmf2deg, cie.illD50);

% put the XYZs through XYZ2dispRGB function to estimate color-calibrated
% RGB 0-255 DCs for the given display
dispRGBs = XYZ2dispRGB(fullfile(paths.p06.data, 'display_model.mat'), ...
    camXYZs, XYZ_D50);

% workflow_test_cal.tif preserves the originally submitted calibrated RGB
% stimuli plus repeated black and white measurements. It is intentionally
% not regenerated from the corrected conversion above.

% load the measured XYZs
cal_XYZs = importdata(fullfile(paths.p07.data, ...
    'workflow_test_cal.tif'), ' ', 20);
% extract the XYZ data for the displayed CC patches
cal_CC_XYZs = cal_XYZs.data(1:24, 5:7)';
% for 3 measurements of display black and average them
cal_XYZk = mean(cal_XYZs.data(25:27, 5:7), 1)';
% for 3 measurements of display white and average them
cal_XYZw = mean(cal_XYZs.data(28:30, 5:7), 1)';

% calculate Lab values for the displayed CC patches from the XYZs
cal_CC_Labs = XYZ2Lab(cal_CC_XYZs, cal_XYZw);

% calculate the delta E to measure the color differences
cal_deltaE = deltaEab(munki_Labs, cal_CC_Labs);

% calculate the min, max and mean delta E values
cal_deltaE_max = max(cal_deltaE);
cal_deltaE_min = min(cal_deltaE);
cal_deltaE_mean = mean(cal_deltaE);

% summarize and display the differences between the real and displayed Lab
% values for the calibrated workflow
printWorkflowError('Calibrated', ...
    ['camera -> RGB_cam -> camera model -> estimated XYZ -> ' ...
    'display model -> RGB_disp -> display'], ...
    munki_Labs, cal_CC_Labs, cal_deltaE);
```

```
% delete temporary vars
clear cie cal_XYZs
```

Calibrated workflow color error
 camera -> RGB_cam -> camera model -> estimated XYZ -> display model -> RGB_disp -> display

Real vs. displayed ColorChecker Lab values							
patch #	real			displayed			dEab
	L	a	b	L	a	b	
1	37.1865	14.9985	15.2592	41.9663	10.6576	13.6162	6.6626
2	65.8188	16.8695	18.0267	68.5009	16.0815	16.4969	3.1867
3	49.9949	-3.1841	-23.5159	52.5802	2.8281	-20.6105	7.1604
4	42.6411	-15.3251	20.0423	44.8993	-14.9659	20.9342	2.4543
5	54.6852	9.6978	-26.7126	55.4015	15.2793	-24.3459	6.1047
6	71.2441	-33.1391	-0.5010	71.3726	-29.9884	-0.1092	3.1776
7	62.2558	34.1094	57.7774	60.1099	33.2074	67.1409	9.6485
8	39.5890	9.9980	-43.6388	41.7823	15.2256	-38.7938	7.4573
9	51.8424	48.1403	16.0636	51.9092	45.8889	16.0172	2.2528
10	29.4495	22.4255	-21.7661	33.4412	24.4566	-20.4213	4.6762
11	71.6264	-24.3441	57.6850	70.8079	-28.3169	60.9746	5.2225
12	72.2288	20.6039	69.0149	71.3460	13.2548	68.2328	7.4431
13	28.6402	18.5907	-51.4092	23.8427	33.6665	-54.6467	16.1486
14	54.6309	-39.5493	32.8341	51.1627	-39.4626	27.5925	6.2858
15	42.5988	54.6049	25.7315	40.0170	55.5741	22.2261	4.4601
16	82.4265	3.8689	78.8570	81.5640	-0.0255	79.4589	4.0339
17	51.5476	49.5154	-14.3758	48.6377	48.9975	-16.1106	3.4272
18	49.3892	-26.5473	-28.6645	48.9482	-18.4414	-24.7165	9.0270
19	95.4458	-0.4414	0.0244	94.9547	-1.5456	2.1532	2.4479
20	80.0339	0.1309	-0.9345	78.8399	-1.3290	0.5130	2.3774
21	66.0107	-0.0004	-1.1463	60.1126	-0.8393	-0.0853	6.0512
22	50.5546	-0.6207	-0.9616	49.3177	-1.2716	0.6662	2.1455
23	35.1532	-0.0632	-0.9708	37.0342	-2.6984	0.2442	3.4581
24	20.3224	-0.2858	-0.5603	13.3120	-6.1799	10.5044	14.3636
min							2.1455
max							16.1486
mean							5.8197

step 3 - visualize the differences between ground-truth, un-calibrated and calibrated renderings of CC chart

```
% load the ground truth XYZ and Lab values of the CC chart
munki_XYZs;
munki_Labs;

% normalize XYZ values to range 0 - 1
munki_XYZs_normalized = munki_XYZs ./ 100;

% use xyz2srgb transform to calculate RGB values from XYZs
CC_RGBs = xyz2srgb(munki_XYZs_normalized, 'WhitePoint', 'd50', 'OutputType', 'uint8');

% create a 4x6x3 array to hold un-calibrated RGBs
camRGBs_255_resized = uint8(reshape(camRGBs_255', [6 4 3]));
camRGBs_255_resized = fliplr(imrotate(camRGBs_255_resized, -90));

% create a 4x6x3 array to hold calibrated RGBs
dispRGBs_resized = uint8(reshape(dispRGBs', [6 4 3]));
dispRGBs_resized = fliplr(imrotate(dispRGBs_resized, -90));

% create a 4x6x3 array to hold ground-truth RGBs
CC_RGBs_resized = uint8(reshape(CC_RGBs, [6 4 3]));
CC_RGBs_resized = fliplr(imrotate(CC_RGBs_resized, -90));

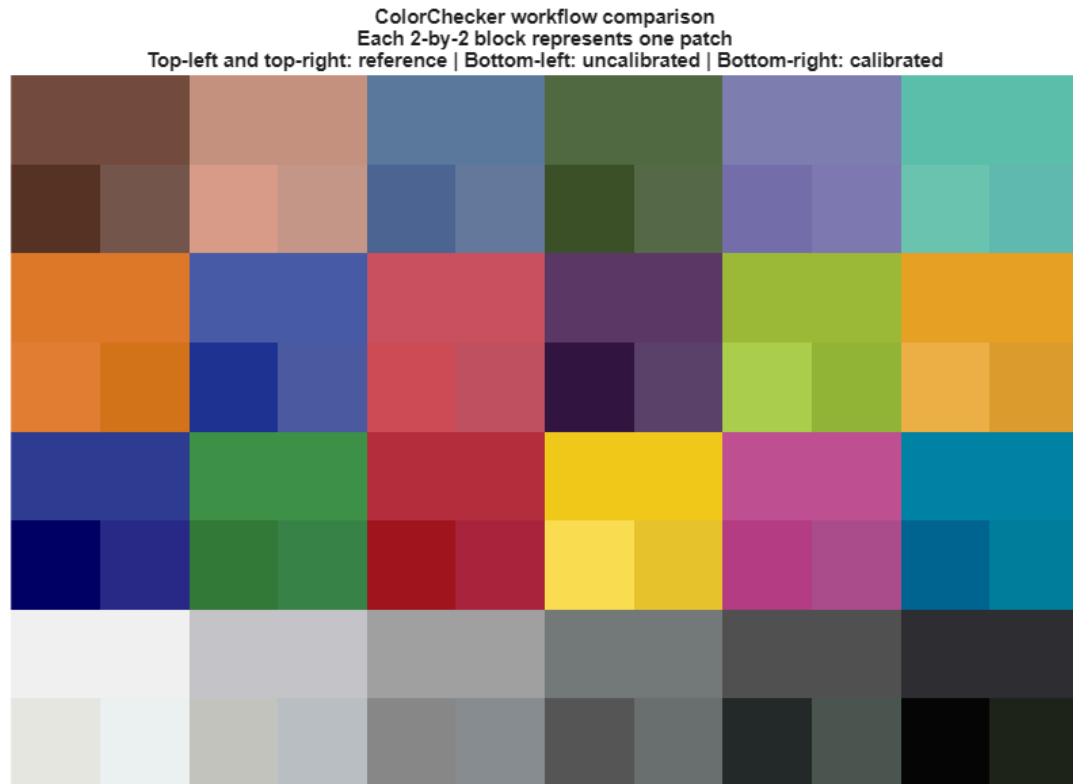
% Interleave ground-truth rows with pairs of uncalibrated/calibrated rows.
workflow_diffs = zeros(8, 12, 3, 'uint8');
workflow_diffs(1:2:end, 1:2:end, :) = CC_RGBs_resized;
workflow_diffs(1:2:end, 2:2:end, :) = CC_RGBs_resized;
workflow_diffs(2:2:end, 1:2:end, :) = camRGBs_255_resized;
workflow_diffs(2:2:end, 2:2:end, :) = dispRGBs_resized;

% visualize the array
figure;
image(uint8(workflow_diffs));
axis image off;
title({'ColorChecker workflow comparison', ...
    'Each 2-by-2 block represents one patch', ...
    ['Top-left and top-right: reference | Bottom-left: uncalibrated | ' ...
    'Bottom-right: calibrated']});

% resize the array to 768 x 1024 x 3
workflow_diffs_resized = imresize3(workflow_diffs, 'OutputSize', [768 1024 3], 'Method', 'nearest');
```

```
% save the resized array as a .png image
imwrite(uint8(workflow_diffs_resized), ...
    fullfile(paths.p07.results, 'final_render_corrected.png'));

clear CC_RGBs_resized camRGBs_255_resized dispRGBs_resized workflow_diffs_resized
```



step 4 - color-accurate imaging

```
% start with the original CC 1125x800 photo
img_orig = imread(fullfile(paths.p07.data, 'original_chart.jpg'));

% reshape the image into a pixel vector/[3 x n] RGB vector
[r, c, p] = size(img_orig);
pix_orig = reshape(img_orig, [r * c, p]);

% process the pixels through camera and display models derived in previous
% projects to calculate color-calibrated DCs
pix_DCs_calib = XYZ2dispRGB( ...
    fullfile(paths.p06.data, 'display_model.mat'), ...
    camRGB2XYZ(fullfile(paths.p05.data, 'cam_model.mat'), ...
        double(pix_orig)), XYZ_D50);

% reshape the pixels back into an 3-d array
img_calib = reshape(pix_DCs_calib, [r, c, p]);

% save the color-calibrated image
imwrite(uint8(img_calib), ...
    fullfile(paths.p07.results, 'color_calibrated_chart_corrected.png'));

clear r c p pix_orig pix_DCs_calib
```

compare the original and color calibrated images

The two images are shown with the same dimensions for a direct visual comparison. The calibrated image is the result of applying the camera model followed by the display model.

```
imageComparisonFigure = figure('Color', 'w', 'Name', ...
    'Original and color-calibrated images');
imageComparisonLayout = tiledlayout(imageComparisonFigure, 1, 2, ...
    'TileSpacing', 'compact', 'Padding', 'compact');

nexttile(imageComparisonLayout);
imshow(img_orig);
title('Original camera image');
```

